Assignment 2: Link Prediction and Ethical Implications

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Abstract. This essay investigates link prediction in social networks using supervised learning, combining structural, local, and community-aware features to predict potential connections between nodes. A Gradient Boosting Classifier was identified as the optimal model, achieving 88% accuracy, with community similarity feature being the most influential predictor. As well as technical performance, the study examines the ethical implications of link prediction, focusing on fairness. Structural biases inherent in social networks can favor well-connected or majority communities, reinforcing existing inequalities. To address these disparities, post-processing methods based on the Equalized Odds were applied, reducing unfairness, but also reducing accuracy by 4%. The essay emphasizes that effective link prediction requires balancing predictive performance with ethical considerations, demonstrating that fairness, interpretability, and social responsibility must be integrated throughout the model design, evaluation, and deployment process.

Keywords: Link Prediction · Gradient boosting · Fairness · Equalized Odds

1 Technical Solution

This section describes a supervised learning solution to the link-prediction task. Given a pair of nodes, predict whether the link between them existed in the original network (1 = existed, 0 = did not exist). This section discusses feature engineering, data preparation and model selection, training and validation. Finally, the results of the model are discussed along with the limitations of the techniques and data used.

1.1 Feature Engineering

To effectively predict whether a link was existent (positive) or non-existent (negative) in the network, both local, global and community aware metrics were explored. For community metrics, an unsupervised community detection approach was applied using greedy modularity, to uncover groupings within the network. The resolution parameter was optimized to maximize modularity, which represents how densely connected the nodes are within these groups. This resulted in 6 communities, as shown in figure 1, with modularity of 0.77.

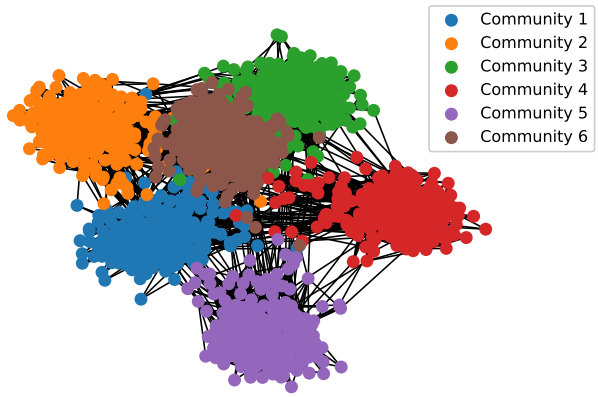

Fig. 1. The 6 communities detected in the graph using Greedy Modularity

Furthermore, several features were initially computed, evaluated on co-authorship networks (Nowell and Kleinberg, 2004). These included resource allocation, Jaccard coefficient, Adamic Adar, preferential attachment, common neighbors, Katz index, clustering coefficients, and indicators on whether the two nodes share the same attribute and belong to the same community created by greedy modularity algorithm. Based on the experimentation on the importance of each feature in the link prediction task, the final model utilized resource allocation index, preferential attachment, the degrees of the nodes, and community similarity indicator. The combination of these measures captures structural information, local connectivity patterns, and also global popularity effects. The use of a small amount of features in the final predictive model was two-fold. The first reason is in regard of technical aspect; some of the measures did not contribute to accurately predicting whether a link exists in the network for any of the selected models. As a result, they were removed to reduce the complexity of the model. However, some features were of minimal importance and could slightly increase the accuracy of the model, such as the attribute similarity metric, but were excluded because they contain sensitive information. Thus, it was decided that a slight trade-off between accuracy and privacy (Bagdasaryan and Shmatikov, 2019) was preferable.

1.2 Data Preparation

The base dataset contains 1500 nodes and 7333 links; approximately 10% of links were removed from the original graph. Firstly, to prepare training data, positive examples were created from existing edges in the graph. Negative examples

were sampled at random from node pairs not present in the original graph while ensuring that sampled negatives did not include any pairs present in the final test input. A balanced positive to negative ratio of 1:1 for the primary training was applied because balanced classes simplify model interpretation and metrics, and also preserves the same balance with the unseen data that the final model would be tested. To validate this method, experiments were conducted under class imbalance conditions, as this graph with 1,500 nodes and an average degree of 5 contains more negative (non-existent) connections than positive ones. Experiments also used Structure Aware Negative Sampling to improve the 'hard' negative mining as suggested by Ahrabian et. al, 2020. However, all of these yielded worse results in training and testing sets than random sampling, therefore due to the benefits in simplicity and performance, random sampling was used.

In addition, the input features were standardized to take into account the significant differences in their ranges. While Pinheiro et al. (2025) found that the performance of ensemble methods such as random forest were unaffected by large differences in range, others such as logistic regression were negatively impacted. In order to conduct model comparisons ensuring the best performance from all models and consistency in the training data the data was scaled.

1.3 Model Selection

To ensure the best model was selected, four classification models were developed and evaluated: Random Forest, Logistic Regression, XGBoost, and Gradient Boosting Classifier, which have proved to be effective in link prediction task (Wang et. al, 2022, Bentéjac et. al, 2019). Based on accuracy performance, Gradient Boosting Classifier slightly outperformed the rest models with the selected features, as discussed in Section 1.1, while the F1 scores were balanced across all models. To optimize model performance, grid search was used to configure optimal hyperparameters for the classifier, including the number of estimators, learning rate, maximum tree depth, and feature sampling strategy.

1.4 Model Training and Validation

The dataset was split into 80% for training and 20% for testing, using stratified sampling to preserve class balance. During grid search, 5-fold cross validation was implemented in 20% of the data for each fold to ensure the robustness of the model by evaluating it on a wider range of data. The parameters were tuned, aiming to maximize accuracy, while also F1 score and ROC-AUC were measured to provide a holistic view of performance. Furthermore, the probability threshold corresponding to the highest accuracy of the model was identified and used to generate the final edge predictions.

1.5 Final Results

The final model achieved 88.8% accuracy in the test set with an F1 score of 88% in both classes. Also, the Area Under Curve score was 93.8%, indicating that

the model can effectively achieve high True Positive Rates (TPR) even at low False Positive Rates (FPR) as shown in figure 2. Additionally, when applied to the unseen prediction dataset, the model achieved an accuracy of 87%, further demonstrating its ability to generalize effectively on unseen data.

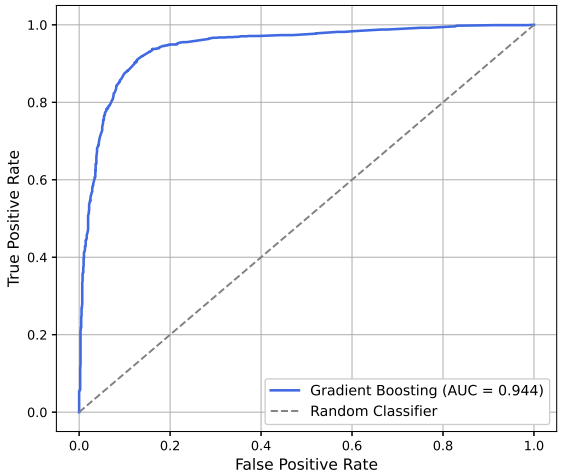

Fig. 2. ROC AUC Curve against a Random Classifier

The feature importance analysis (Table. 1) provides further information on the drivers of the model. The indicator on whether the two nodes belong in the same community is the most significant predictor, suggesting that a community aware metric is the most important feature in this particular link prediction task, compared to the rest local structure and global popularity metrics.

Table 1. Feature Importance scores of Gradient Boosting Classifier model

Feature	Score
community similarity	67.8
resource allocation	19.1
degrees sum	9.3
preferential attachment	3.8

1.6 Limitations

Although the final model effectively addressed the link prediction task with high accuracy, several limitations must be acknowledged. While feature importance analysis can provide useful information on the underlying factors influencing the model's predictive behavior, a Gradient Boosting Classifier model does not provide sufficient interpretability at the level of individual predictions. Furthermore, while the use of a small number of features helps avoid the inclusion of sensitive information, the use of a larger number of features could enhance the model's ability to capture complex structural dependencies and patterns within the network. Future work could focus on integrating explainable AI techniques within Graph Neural Networks (Zhu et al., 2023), which have also demonstrated superior performance in handling complex network structures in most settings (Li et al., 2023). Finally, the dataset used for training and testing was balanced with a 1:1 ratio of positive to negative edges, aligning with the composition of the unseen prediction dataset. However, in real-world networks, the vast majority of possible node pairs are not connected, leading to a highly imbalanced class distribution where negative links far outnumber positive ones. Consequently, the model's performance may not be as effective in larger, sparse networks.

2 Ethics in Link Prediction

2.1 Introduction

The supervised link prediction model developed in 1 Technical solution aims to predict potential connections between individuals in a social network by learning structural patterns such as shared communities and mutual neighbors. While this approach achieved predictive accuracy of 88.8%, it also raises significant ethical questions. In social networks, model predictions are not merely computational outcomes but instead shape online behaviors, control who people engage with, what communities they join, even the information that they are exposed to. As a result, the deployment of link prediction systems demands close attention to three different ethical dimensions that are widely discussed in the field of data science: fairness, privacy, and transparency (Floridi & Taddeo, 2016; Mittelstadt et al., 2016). Fairness concerns arise when models reproduce or amplify existing inequalities embedded in network data, favouring well-connected users while dismissing more isolated ones (Friedman & Nissenbaum, 1996; Barocas et al. 2023). Privacy issues emerge because link prediction can deduce relationships between users, while some of them being undisclosed or highly sensitive (Nissenbaum, 2009). Lastly, transparency poses an additional challenge, as users don't know the algorithmic process behind link prediction, thus limiting their ability to explain why and how certain connections are being recommended (Doshi-Velez & Kim, 2017). Among these concerns, fairness constitutes the most significant issue for the developed model. As the classifier relies heavily on community features, it may systematically privilege majority clusters and overlook minority or less connected users, reinforcing structural disparities already present in the network. The following section therefore examines the concept of fairness in link prediction in depth, analysing its underlying causes, social implications, and potential strategies for mitigation.

2.2 Elaboration

What is Fairness in Link Prediction In the context of link prediction, fairness ensures that the model identifies patterns in a way that treats individuals and groups equally in its predictions. Fairness can be examined under two complementary perspectives: individual fairness and group fairness (Dwork et al., 2012). While individual fairness argues that similar users with comparable profiles and behaviors should receive similar link predictions, group fairness states that groups with different demographics and attributes should have similar probabilities of receiving link recommendations.

In addition, fairness in the context of link prediction can be related to formal metrics such as Demographic Parity and Equalized Odds. The former affirms that each group should have an equal rate of positive link predictions and ensures that no group systematically receives more or fewer positive link predictions. The latter requires models to predict similar false positive and negative rates across groups, aiming to have proportional prediction errors (Hardt, Price, & Srebro,

2016). These fairness principles help to measure and evaluate the ethical aspects of link prediction systems, especially when sensitive or group-based features are used when training the model. In our case, the link prediction model exhibited an Equalized Odds difference of 0.197, indicating a disparity of nearly 20% between the most and least favored communities in terms of true and false positive rates. This means that the model's accuracy in predicting links varies significantly across different groups, thereby violating the Equalized Odds criterion. In practical terms, certain communities are systematically more likely to have their existing or potential connections correctly identified, while others are more prone to being misclassified.

Bias Types and Their Origins The main reason for unfairness in link prediction tasks mostly stems from the structure of social networks and the way the model tries to identify patterns and learn from them. In social networks there is a tendency for individuals to be associated with others with similar backgrounds such as ethnicity, gender, interests, or socioeconomic levels (McPherson, Smith-Lovin, & Cook, 2001). In other words, the majority groups tend to have dense connections, while the minority groups are linked with fewer.

Our approach leverages several structural and connectivity features such as resource allocation, preferential attachment, degrees sum, and the community similarity indicator. Even though these features represent social network dynamics effectively, they inherently favor users who already have a large number of connections, which reinforces the visibility of popular users while disadvantaging others. Therefore, individuals who have more connections are more likely to be predicted positively and suggested with more connections, while isolated or minority users tend to get recommended less, which can be referred to as the rich-get-richer dynamic (Barabási & Albert, 1999). The community similarity feature has been proved to be the strongest predictor in our model; however, since it primarily recommends connections within similar groups, it risks promoting existing social groups. As a result, the model may compromise fairness despite having high evaluation scores by inherently reducing diversity, and providing unequal exposure and interactions.

Demographic Bias In social networks, even if sensitive attributes such as gender, race, or political opinion are excluded from the model, node attributes and graph structures often correlate with demographic or cultural traits. This may cause proxy discrimination, where neutral features produce outcomes that are unfair to specific demographic groups (Zemel et al., 2013).

These structural biases are also reflected in the first part's model predictions, as illustrated in Figure 3. The figure presents the True Positive Rate (TPR) and the False Positive Rate (FPR) for each community from the model in Technical solution 1. While the TPR remains relatively high and the FPR low across most communities, the selection rates vary noticeably. This suggests that certain communities receive positive predictions less frequently, despite comparable accuracy levels. Such disparities reveal how the model's reliance on degree-based

and community features can reproduce existing structural inequalities within the network.

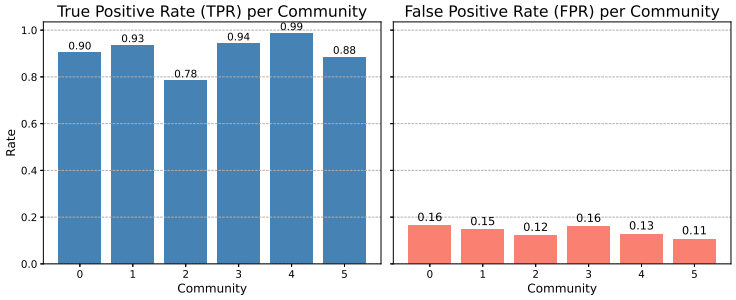

Fig. 3. True Positive Rate and False Positive Rate per community

Fairness Metrics and Trade-Offs Fairness metrics cannot perfectly identify unfairness or exactly who or which group is being disadvantaged. Additionally, some metrics often conflict with each other. For instance, demographic parity cannot detect structural differences in social networks if one group has fewer connections due to being a small community. Even though equalized odds is considered more robust, it still requires an unbiased ground truth, which is almost impossible since the training data themselves are already naturally biased. The trade-off between these metrics causes fairness dilemmas in other domains. The COMPAS re-offender prediction case, where the model achieved similar accuracy scores across racial groups but showed significantly higher false positive rates for Black defendants, is a prominent example of this (Angwin et al., 2016).

2.3 Mitigation

Ensuring fairness in link prediction models requires interventions throughout the data science cycle, from how data is prepared to how models are trained and how their outputs are interpreted. Bias can originate from different sources like pre-existing social structures, model architecture choices and technical limitations, or emergent use contexts (Friedman & Nissenbaum, 1996). Consequently, fairness mitigation can take place at pre-processing, in-processing, and post-processing. This layered approach highlights that promoting fairness is not a single technical fix but a process that must be embedded across all stages of model development and evaluation.

Pre-processing Pre-processing techniques address bias by transforming the data to ensure that learning is happening to unbiased data sets. Since unfairness may occur from the data itself, methods that fall into this category aim

to correct imbalances prior to model training. Two main approaches are re-weighting and re-sampling (Kamiran & Calders, 2012). Re-weighting adjusts the importance of individual samples so that underrepresented groups have a proportionally stronger influence on the model's learning process. Even though these interventions make the data more balanced of equitable conditions, excessive weighting or sampling can reduce model accuracy and generalizability. In our case, pre-processing methods were not applied, as altering the data could have disrupted the structural properties of the network and introduced artificial biases, especially since the dataset lacked clearly defined sensitive attributes beyond community labels.

In-processing In-processing methods integrate fairness constraints directly into the model's training process, modifying the algorithm's learning objective to take into account biased behavior. Instead of manipulating the data, these approaches introduce a fairness term into the loss function to penalize discrimination between groups. Together they minimize prediction error and group level unfairness (Kamishima et al., 2012). For example, fairness objectives based on Equalized Odds or Equality of Opportunity can be incorporated into the optimization process to ensure that false positive and false negative rates are approximately the same across different groups (Hardt et al., 2016). Likewise, no in-processing methods were applied, as the model already achieved accuracy of 88.8%, which served as the performance benchmark. Introducing fairness constraints during training could have required retraining and risked lowering this baseline accuracy, so instead post-processing techniques were implemented that preserved the model's learned parameters while addressing fairness at the output level.

Post-Processing Post-processing methods change predictions or decision thresholds without interfering with the data or the training of the model in order to mitigate unfair outcomes (Hardt et al. ,2016). Rather than re-training the model with different parameters, the decision boundaries are adjusted between different groups to balance False Positive Rate (FPR) and False Negative Rate (FNR). To accomplish that, fairness metrics are introduced such as Equalised Odds or Demographic Parity that guide the shift of the threshold for unrepresented groups (Hardt et al. ,2016). While these methods are simple to implement as it only requires interventions on already deployed systems, they cannot fix deeper biases on the raw data (Friedman & Nissenbaum, 1996; Dobbe et al., 2018).

For this study, fairness mitigation was applied exclusively at the post-processing stage, where group-specific decision thresholds were optimized according to the Equalized Odds criterion to balance true and false positive rates across the six communities. This approach allows fairness constraints to be enforced without retraining the model or altering the underlying data. The Equalized Odds difference of the base model was 0.20, which decreased to 0.092 after post-processing, corresponding to a 52% reduction in disparity. The results, summarized in Table 2, show markedly more consistent True Positive Rate (TPR) and False Posi-

tive Rate (FPR) values across communities, indicating that the post-processing intervention effectively reduced unequal treatment between network groups.

Table 2. True Positive Rates and False Positive Rates per Community before and after applying Equalised Odds

Community	Base model TPR	Base model FPR	Fair model TPR	Fair model FPR
0	0.903	0.163	0.806	0.168
1	0.934	0.145	0.821	0.115
2	0.783	0.121	0.787	0.116
3	0.942	0.159	0.792	0.139
4	0.985	0.127	0.855	0.136
5	0.884	0.106	0.884	0.111

Nevertheless, this improvement in fairness was accompanied by a small drop in predictive accuracy, from 88% to 84%, illustrating the trade-off between fairness and performance. This trade-off arises because enforcing parity in model outcomes often requires relaxing thresholds that previously maximized accuracy (Hardt, Price, & Srebro, 2016). While post-processing provides a practical and minimally invasive means of enhancing fairness in deployed systems, it operates only at the output level and cannot address structural or data-driven biases that may persist in the model's representations. Consequently, post-processing should be viewed as a complementary fairness intervention, to be integrated alongside pre-processing and in-processing techniques, in order to achieve a more ethically and socially responsible link prediction model.

2.4 Conclusion

This essay has explored fairness as a central ethical concern in link prediction within social networks, demonstrating that predictive accuracy alone cannot guarantee fair outcomes. By examining how structural features such as degree centrality and community similarity shape algorithmic decisions, it becomes clear that link prediction models tend to reinforce existing social hierarchies. The analysis further emphasized that fairness can be pursued throughout the data science process, intervening either before, after or while the model is being trained, taking into account inevitable trade-offs between fairness and accuracy. In the specific network, the post-processing adjustments effectively reduced disparities between groups without the need for retraining the model. However, as equality across all groups was not fully achieved, the results underscore that meaningful bias mitigation must be integrated throughout the entire data science lifecycle.

Ensuring fairness in link prediction requires a comprehensive approach that considers moral and social values at every stage of model design and deployment. As link prediction algorithms increasingly shape peoples interactions, the challenge for data scientists is not merely to minimize statistical disparities but to cultivate systems that uphold social justice, accountability, and inclusivity.

References

1. Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). *Machine bias*. ProPublica. https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing
2. Barabási, A.-L., & Albert, R. (1999). Emergence of scaling in random networks. *Science, 286*(5439), 509–512. https://doi.org/10.1126/science.286.5439.509
3. Barocas, S., Hardt, M., & Narayanan, A. (2019). *Fairness and machine learning*. Online draft. https://fairmlbook.org/
4. Binns, R. (2018). Fairness in machine learning: Lessons from political philosophy. In *Proceedings of the 2018 Conference on Fairness, Accountability, and Transparency* (pp. 149–159). ACM. https://doi.org/10.1145/3287560.3287586
5. Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2012). Fairness through awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference* (pp. 214–226). ACM. https://doi.org/10.1145/2090236.2090255
6. Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems* (pp. 3315–3323).
7. McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology, 27*(1), 415–444. https://doi.org/10.1146/annurev.soc.27.1.415
8. Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys, 54*(6), 1–35. https://doi.org/10.1145/3457607
9. O'Neil, C. (2016). *Weapons of math destruction: How big data increases inequality and threatens democracy*. Crown Publishing Group.
10. Ahrabian, K., Feizi, A., Salehi, Y., Hamilton, W. L., & Bose, A. J. (2020). Structure aware negative sampling in knowledge graphs. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 6093–6101). Association for Computational Linguistics. https://aclanthology.org/2020.emnlp-main.492
11. Pinheiro, J. M. H., Oliveira, S. V. B., Silva, T. H. S., Saraiva, P. A. R., Souza, E. F., Ambrosio, L. A., & Becker, M. (2025). The impact of feature scaling in machine learning: Effects on regression and classification tasks. *arXiv preprint* arXiv:2506.08274v2. https://arxiv.org/abs/2506.08274v2
12. Bentéjac, C., Csörgő, A., & Martínez-Muñoz, G. (2019). A comparative analysis of XGBoost. *arXiv preprint* arXiv:1911.01914 [cs.LG]. https://arxiv.org/abs/1911.01914
13. Bagdasaryan, E., & Shmatikov, V. (2019). Differential privacy has disparate impact on model accuracy. *arXiv preprint arXiv:1905.12101*. https://doi.org/10.48550/arXiv.1905.12101
14. Liben-Nowell, D., & Kleinberg, J. (2003). The link prediction problem for social networks. In *Proceedings of the Twelfth International Conference on Information and Knowledge Management (CIKM '03)* (pp. 556–559). Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/956863.956972
15. Li, J., Shomer, H., Mao, H., Zeng, S., Ma, Y., Shah, N., Tang, J., & Yin, D. (2023). Evaluating graph neural networks for link prediction: Current pitfalls and new benchmarking. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS '23)*, Article 169, 3853–3866. Curran Associates Inc., Red Hook, NY, USA.

16. Zemel, R., Wu, Y., Swersky, K., Pitassi, T., & Dwork, C. (2013). Learning fair representations. In *Proceedings of the 30th International Conference on Machine Learning (ICML '13)* (pp. 325–333). JMLR.org. https://proceedings.mlr.press/v28/zemel13.html
17. Zhu, H., Luo, D., Tang, X., Xu, J., Liu, H., & Wang, S. (2023). Self-explainable graph neural networks for link prediction. *arXiv preprint arXiv:2305.12578*. https://arxiv.org/abs/2305.12578
18. Wang, T., Jiao, M., & Wang, X. (2022). Link prediction in complex networks using recursive feature elimination and stacking ensemble learning. *Entropy, 24*(8), 1124. https://doi.org/10.3390/e24081124
19. Friedman, B., & Nissenbaum, H. (1996). Bias in computer systems. *ACM Transactions on Information Systems (TOIS), 14*(3), 330–347. https://doi.org/10.1145/230538.230561
20. Kamiran, F., & Calders, T. (2012). Data preprocessing techniques for classification without discrimination. *Knowledge and Information Systems, 33*(1), 1–33. https://doi.org/10.1007/s10115-011-0463-8
21. Kamishima, T., Akaho, S., Asoh, H., & Sakuma, J. (2012). Fairness-aware classifier with prejudice remover regularizer. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases* (pp. 35–50). Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-642-33486-3_3
22. Dobbe, R., Dean, S., Gilbert, T. K., & Keel, J. (2018). A broader view on bias in automated decision-making: Reflecting on epistemology and dynamics. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 880–890. https://doi.org/10.1145/3442188.3445918